PHASE 1 | NETWORKING | PART 3

IP Addressing

Subnetting & CIDR

IPv4 Structure · Public vs Private · RFC 1918 · Subnet Mask · CIDR · ipcalc · Nmap

IPv4	RFC 1918	/24 CIDR	Subnetting	ipcalc	nmap -sn

AppSec Context

Recon → identify target IPs · CIDR scan for live hosts · Network segmentation = defense · Internal enum after pivot

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What is an IP Address?

DEFINITION

Internet Protocol Address — a unique logical identifier assigned to every device on a network.

Think of it as a postal address for your device. Without it, packets have nowhere to go.

IPv4 STRUCTURE — 32 bits / 4 Octets

192	168	1	10
11000000	10101000	00000001	00001010
Network	Network	Network	Host

Each octet: 8 bits → value range: 0 – 255

Total = 2^{32} = ~4.3 Billion addresses

IPv4

32-bit · 4.3B addresses · Still dominant
Format: 192.168.1.10

IPv6

128-bit · 340 undecillion addresses · Growing
Format: 2001:0db8::1

Public vs Private IP

PRIVATE IP RANGES (RFC 1918)

Not routable on the internet. Internal use only.

/8
`10.0.0.0 — 10.255.255.255`
Large enterprises / ISP · 16.7M hosts

/12
`172.16.0.0 — 172.31.255.255`
Medium networks · 1M hosts

/16
`192.168.0.0 — 192.168.255.255`
Home / Small office · 65K hosts

PUBLIC IP

Assigned by your ISP. Visible on the internet. Unique globally.

This is your attack surface.

`cmd: curl ifconfig.me`

SPECIAL ADDRESSES

- `127.0.0.1` **Loopback**
  Your own machine. If this fails — network stack broken.
- `0.0.0.0` **Default Route**
  'Any destination.' Used in routing tables.
- `255.255.255.255` **Broadcast**
  Send to all hosts on segment simultaneously.

RECON CONTEXT

Public IP found → attack surface mapped. Private range found (post-pivot) → internal enumeration begins. nmap -sn 10.0.0.0/8

IP Practical — Kali Terminal

Run these on your Kali machine. Understand what every line means.

```
$ ip addr show
```

WHAT IT DOES

Shows all network interfaces and their assigned IPs.

LOOK FOR

eth0 = main interface. lo = loopback (127.0.0.1). inet line = your IPv4 address + CIDR.

```
$ ping <metasploitable-ip>
```

WHAT IT DOES

Test ICMP connectivity to target VM.

LOOK FOR

Reply = machine is reachable. No reply = firewall blocking ICMP or host is down.

```
$ ping 127.0.0.1
```

WHAT IT DOES

Loopback test — are you pinging yourself?

LOOK FOR

Reply = network stack is alive. Request timeout = something is broken at OS level.

```
$ ip addr show eth0
```

WHAT IT DOES

Show only eth0 interface detail.

LOOK FOR

Look at inet line → your IP in CIDR format e.g. 192.168.1.100/24 ← that /24 is subnetting.

What is Subnetting?

DEFINITION

Dividing a large IP network into smaller sub-networks. Each subnet is an isolated broadcast domain.

Reason 1 — IP conservation: Don't waste 246 addresses if you only need 10.

Reason 2 — SECURITY:

Contain a breach. If attacker compromises Subnet A — Subnets B and C stay isolated. This is Network Segmentation.

SUBNET MASK — How it works

IP Address	192	168	1	10
Subnet Mask	255	255	255	0
Binary	11111111	11111111	11111111	00000000

← NETWORK PORTION (locked)

HOST PORTION →

RULE:

255 in subnet mask = network bits (fixed). 0 = host bits (variable). First 3 octets = network. Last octet = devices on that network.

CIDR Notation

CIDR — Classless Inter-Domain Routing (IETF, 1993)

Replaced the rigid Class A/B/C system. Notation: IP/prefix-length where prefix = number of network bits.

FORMULA

```
Total addresses  =  2^(32 - prefix)
Usable hosts     =  2^(32 - prefix)  -  2
```

(subtract network addr + broadcast addr)

EXAMPLE — 192.168.1.0/24

Prefix /24 → 24 bits = network

8 bits left for hosts → 2^8 = 256 total

256 – 2 = 254 usable hosts

CIDR REFERENCE TABLE

CIDR	Subnet Mask	Total IPs	Usable Hosts	Use Case
/8	255.0.0.0	16.7M	16,777,214	Large ISPs, enterprise
/16	255.255.0.0	65K	65,534	Campus / data center networks
/24	255.255.255.0	256	254	Typical LAN / home network
/28	255.255.255.240	16	14	Small segments / VLAN
/30	255.255.255.252	4	2	Point-to-point router links
/32	255.255.255.255	1	0 (host)	Single host route

Subnetting Practical — Lab

ipcalc + nmap on Kali against Metasploitable 2

```
$ ipcalc 192.168.1.0/24
```

Subnet calculator — breaks down any CIDR block completely.

```
Network: 192.168.1.0
HostMin: 192.168.1.1
HostMax: 192.168.1.254
Broadcast: 192.168.1.255
Hosts/Net: 254
```

```
$ ipcalc 10.0.0.0/8
```

Large enterprise/ISP range — see the scale difference.

```
Network: 10.0.0.0
HostMin: 10.0.0.1
HostMax: 10.255.255.254
Broadcast: 10.255.255.255
Hosts/Net: 16,777,214
```

```
$ ipcalc 192.168.1.0/28
```

/28 — much smaller block. Compare output with /24.

```
Network: 192.168.1.0
HostMin: 192.168.1.1
HostMax: 192.168.1.14
Broadcast: 192.168.1.15
Hosts/Net: 14
```

```
$ nmap -sn 192.168.1.0/24
```

Ping scan — discover all live hosts in subnet. No port scan.

```
-sn = no port scan (ping only)
Finds Metasploitable 2 IP
ICMP / ARP used for discovery
First step of every pentest
Report: hosts up / down
```

Key Takeaways

> "If you cannot read the network, you cannot attack it. If you cannot attack it, you cannot defend it."

01 **IPv4 = 32 bits / 4 octets**

Value: 0–255 per octet. Total: ~4.3B addresses. Format: 192.168.1.10

02 **Private ranges (RFC 1918)**

10.x, 172.16–31.x, 192.168.x — not routable on internet. Your lab network.

03 **Subnetting = network division**

Subnet mask separates network bits from host bits. Security benefit: segmentation.

04 **CIDR /24 = 254 usable hosts**

Formula: 2^(32-prefix) − 2. /30 = 2 hosts. /32 = single host.

05 **ipcalc + nmap -sn**

ipcalc breaks down any CIDR. nmap -sn finds all live hosts — recon step 1.

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#IPAddressing #Subnetting #CIDR #NetworkSecurity #Cybersecurity #LearningInPublic